Society of Petroleum Engineers

SPE-229189-MS

Video Analytics for Workplace Safety

Marco Memmoli and Lorenzo Corallo, Saipem SpA, Fano (PU), Italy; Jacopo Moretti, Saipem SpA, Milano (MI), Italy

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Industrial operations in the energy and construction sectors have been reshaped in recent years by the accelerated integration of digital technologies and advanced data-driven methodologies into safety management. As projects grow in complexity, scale, and geographical reach, organizations are increasingly turning to innovative technological solutions to supplement and enhance the effectiveness of their existing HSE strategies.

In this evolving landscape, Video Analytics systems powered by artificial intelligence, specifically computer vision, and analytics platforms have become pivotal in transforming the way unsafe acts and unsafe conditions are detected, monitored, and managed across a wide range of operational environments. These technologies are being deployed not in isolation, but as part of a holistic approach that builds upon the enduring strengths of traditional HSE frameworks.

The synergy between technological innovation and established safety methodologies is empowering teams to manage risks more effectively, ensure compliance with operational standards, and reinforce a culture of safety throughout the organization.

Introduction

Industrial operations in the energy and construction sectors have been reshaped in recent years by the accelerated integration of digital technologies and advanced data-driven methodologies into safety management. As projects grow in complexity, scale, and geographical reach, organizations are increasingly turning to innovative technological solutions to supplement and enhance the effectiveness of their existing HSE strategies.

In this evolving landscape, Video Analytics systems powered by artificial intelligence, specifically computer vision, and analytics platforms have become pivotal in transforming the way unsafe acts and unsafe conditions are detected, monitored, and managed across a wide range of operational environments. These technologies are being deployed not in isolation, but as part of a holistic approach that builds upon the enduring strengths of traditional HSE frameworks.

The synergy between technological innovation and established safety methodologies is empowering teams to manage risks more effectively, ensure compliance with operational standards, and reinforce a culture of safety throughout the organization.

Local Video Analytics Platform: AI in the Field

Central to the implementation of the Video Analytics system is a sophisticated, AI-driven platform purpose-built to operate seamlessly in both offshore and onshore environment. The solution has been already rolled out and implemented across a range of operational contexts, including:

- Six drilling vessels
- One construction vessel
- One onshore site

This broad deployment demonstrates the system's versatility and its ability to adapt to the unique requirements of both offshore and onshore environments. Each environment presents a unique mix of risk factors, spatial configurations, and workflow dynamics that shape both the technical and procedural requirements of safety monitoring. The process begins with a careful mapping of safety-critical zones, areas where the likelihood and potential consequences of unsafe behavior are highest. Dedicated cameras are strategically installed to provide broad and redundant coverage, minimizing the risk of visual occlusion while accommodating constraints imposed by site layout, lighting, and operational flow. Camera selection and placement are never generic: they are defined in close collaboration with field teams and technical experts, ensuring that every angle is purposeful and aligned with real-world risk scenarios. Particular attention is given to minimizing blind spots, considering also how site layout and ongoing construction activities evolve over time.

Data acquisition is only the first step in a much larger pipeline. Video feeds are continuously processed, whose architecture has been adapted and refined through iterative training cycles involving extensive domain expertise. Annotated datasets are developed using real examples from the portfolio of operations, ensuring that each model "understands" the subtleties of equipment, uniforms, PPE, lighting conditions, and operational patterns specific to different sites.

Figure 1—illustrates the annotation protocol applied at frame level. In this example the label list includes helmet, coverall, and person.

A defining strength of this approach is its commitment to site-specific adaptability. Each AI model is an evolving tool that incorporates new data and feedback as conditions change. For example, helmet detection models are trained not only to recognize standard-issue PPE but also to cope with partial occlusion, fast movement, and cluttered scenes typical of active work areas. Similarly, coverall compliance models are fine-tuned to distinguish the nuances between compliant and non-compliant attire, even when visibility is reduced or workers are in non- standard positions. The models supporting exclusion zone and red zone monitoring violations are dynamically linked to site-specific geofences and operational schedules, ensuring that their output remains relevant as workflows and layouts evolve.

The scope of use cases reflects both breadth and maturity. In addition to helmet and coverall detection, exclusion zones, and red zone monitoring, the system can identify unsafe proximity between workers and moving machinery, monitoring personnel presence under suspended loads, detecting spill and leak events, and flagging housekeeping violations such as obstructed walkways or misplaced materials. The pipeline for each model includes starts with the HSE gathering requirements, then continuous data collection, augmentation and retraining, leveraging supervisor annotations and real-world corrections and tuning to improve the overall accuracy of each model and resilience over time.

Figure 2—Example of PPE non-compliance (no-helmet) detected in an offshore construction scenario (PTZ camera)

Figure 3—Example of PPE non-compliance (no-coverall) detected in an offshore construction scenario (PTZ camera)

Figure 4—Example of spill (slippery surface) detected in an offshore construction scenario (PTZ camera)

Figure 5—Example of Poor Housekeeping (Overflowing Bins) detected in an offshore construction scenario (PTZ camera)

Figure 6—Example of exclusion zone (no-entry zone) on an offshore drilling vessel (Fixed Camera)

Figure 7—Example of Red Zone Monitoring use case on an offshore drilling vessel (Fixed Camera)

Operational integration is achieved through a user-centric local dashboard that acts as the center for HSE supervision. Supervisors and HSE personnel in charge are notified in near real time when the system detects a potential unsafe act or unsafe condition. Each event is presented with supporting frame and contextual information, allowing supervisors to quickly assess, validate, or dismiss the AI detection. This workflow not only ensures data quality, but also reinforces the authority of site teams, embedding the technology as a natural extension of the existing safety culture rather than a replacement.

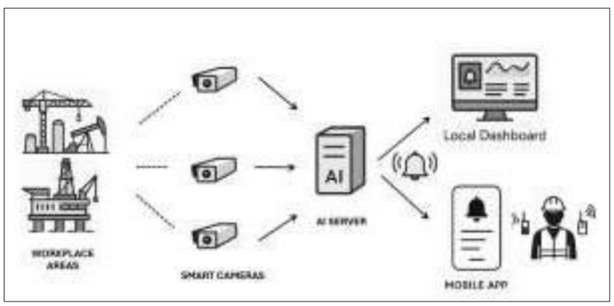

Figure 8—Communication Flow

A critical insight gained from field deployment is the power of iterative, supervisor-led feedback. Rather than operating as a "black box," the AI system is intentionally open to human correction. Dedicated users regularly annotate false positives, add contextual clarifications, or suggest refinements to event boundaries and detection thresholds. This collaborative dynamic accelerates the system's learning curve, mitigates the risk of model drift, and builds a culture of trust between operational staff and the digital tools that support them. Notably, this approach has led to demonstrable improvements in both detection performance and user engagement.

The local dashboard for each asset where Video Analytics has been deployed also serves as a platform for continuous improvement. Analytical views, such as time series of unsafe events, enable supervisors to identify emerging trends, diagnose recurrent hazards, and proactively adapt intervention strategies. Mobile integration further extends the reach of the system, ensuring that even teams working remotely or in challenging environments remain connected to the flow of safety intelligence. In this way, the video analytics platform functions not merely as a passive surveillance tool, but as an enabler of smarter and more responsive supervision that directly supports site leaders.

A key rationale for this dual interface, dashboard and mobile application, is to shorten the loop between detection and action. The system is designed to accelerate the recognition of unsafe acts or conditions and reduce response time, thereby translating situational awareness into faster, more effective interventions

Critical to the success of the video analytics initiative has been the rigorous management of expectations at every organizational level. While the capabilities of artificial intelligence are considerable, the narrative consistently conveyed to all stakeholders has emphasized that technology is an enabler, never a substitute, for professional judgment, field experience, and procedural rigor. Training sessions and onboarding workshops have therefore included not only hands-on demonstrations of the system's functionality but also open discussions about its strengths and limitations. This transparent approach has fostered a sense of ownership and agency among HSE teams, who remain at the center of the decision-making process.

As deployment has expanded from initial pilot site to a broader range of operational sites, the process of knowledge transfer and continuous improvement has become increasingly important. Local actors, often HSE coordinators or safety supervisors, have played a pivotal role in facilitating adoption, sharing practical insights, and acting as bridges between the operational field and the central technical teams. Lessons learned from early deployments have informed refinements to both the AI models and the user interfaces, ensuring that the platform evolves in step with operational realities rather than in isolation from them.

A notable dimension of the deployment process is the focus on environmental adaptation. Each site, whether offshore or onshore, presents unique visual and operational characteristics that may impact detection accuracy and system performance. One of the most significant challenges encountered is the phenomenon known as domain shift, whereby changes in background, lighting, equipment types, or workforce behaviors across different operational environments can lead to a drop in model accuracy if not proactively addressed. Accordingly, the AI training pipeline incorporates cycles of site-specific data collection, targeted annotation, and adaptive model retraining, specifically designed to mitigate the effects of domain shift and ensure robust generalization. This enables the system to maintain high reliability despite variations in lighting, equipment configuration, and workflow dynamics.

Field feedback, such as the identification of zones with persistent false positives and false negatives, is incorporated into the AI development roadmap, supporting agile, iterative refinement. In practice, this means that the video analytics platform is never static, but continuously improves as more data, feedback, and operational context are integrated into its design, ensuring ongoing resilience against domain shift and other real-world challenges.

From Local Validation to Global Insight: Power BI Dashboard

While the local video analytics system is indispensable for immediate detection and response, its true strategic value is realized when locally validated events are aggregated, structured, and analyzed.

The Global Dashboard, developed on the Power BI platform, serves as the central hub for aggregating data. It is here that the individual data points generated daily by site-level AI systems are transformed into actionable knowledge for decision-makers at every tier.

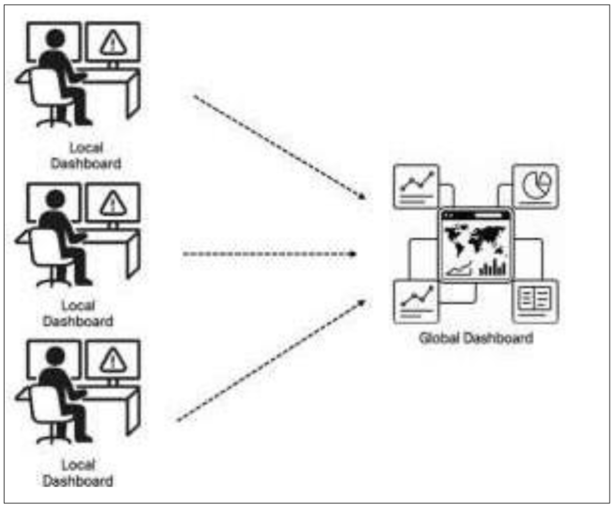

Figure 9—From Local to Global Dashboard

The flow from local to global is meticulously designed for integrity and transparency. Only events that have been reviewed and validated like an unsafe act or condition (generally true positives) by supervisors at site level are eligible for aggregation in the Global Dashboard.

Each unsafe event is transmitted, event type, timestamp, site and area, and camera source, ensuring that every data point is both traceable and meaningful in its operational context.

Interactive visualizations present a panoramic view of unsafe event trends, disaggregated by site, event type, time period, and operational area. Frequency distributions, and time series analyses provide clear signals on where AI observations are most concentrated, how it evolves over time, and where interventions are likely to be most effective.

Key indicators include the total number of unsafe events detected, the most frequent event types, site and area rankings by event frequency, and records for follow-up analysis. Supervisors, managers, and executives can access customized views filtered by project and site type, ensuring that information remains both relevant and actionable for every organizational level.

Crucially, the Global Dashboard is not a passive repository, but a catalyst for organizational learning and strategic alignment. The structured analytics empower not only compliance, but a proactive, prevention-oriented mindset that is essential for achieving step-change improvements in safety performance.

In parallel, the governance of the Global Dashboard has evolved to reflect the growing sophistication of enterprise- wide safety intelligence. Rather than functioning as a one-way aggregator, the dashboard serves as an interactive platform that supports structured dialogue between field operations and corporate leadership. The system enables HSE managers to drill down from macro-level trends to granular, unsafe acts and conditions narratives, allowing for detailed reviews, cross-site benchmarking, and the identification of both best practices and emerging unsafe acts and conditions.

A significant advantage of the dashboard lies in its support for dynamic, data-driven resource allocation. By continuously monitoring event frequencies across multiple sites and operational contexts, leadership can deploy targeted interventions, such as focused training, in a timely and efficient manner. The ability to track the impact of these interventions over time, as reflected in subsequent trends and analytics, closes the loop between action and outcome, ensuring that the organization remains agile and accountable.

Data Governance, Security, and Assurance

In parallel with the functionality, the project team has analyzed several other aspects. In fact, Artificial intelligence solutions, such as Video Analytics, pose significant challenges for the protection of personal data and privacy rights. Such solutions may involve the processing of sensitive information, such as biometric data, behavioral patterns, or location data, which require adequate safeguards and compliance with the applicable legal framework. Moreover, artificial intelligence solutions may raise ethical and social issues, such as transparency, accountability, and human dignity, which need to be addressed with appropriate measures and policies. Therefore, the management of privacy aspects related to artificial intelligence solutions requires a comprehensive and multidisciplinary approach, involving technical, legal, and organizational aspects, as well as the involvement of relevant stakeholders and the respect of human rights principles.

Cyber security as well is a crucial aspect of any artificial intelligence (AI) system, especially when it is applied to sensitive domains such as energy industry. AI systems can be vulnerable to various types of attacks, such as data poisoning, adversarial examples, model stealing, or backdoor insertion, that can compromise their integrity, availability, or confidentiality. Therefore, managing cyber security aspects related to AI requires a comprehensive and proactive approach that involves multiple stakeholders and levels of intervention. The VAWS analysis team has taken the following steps to enhance the cyber security of AI systems:

- Adopting secure design principles and best practices for developing, testing, and deploying AI systems, such as following security standards, frameworks, and guidelines, implementing secure coding techniques, conducting regular security audits and assessments, and applying encryption and authentication mechanisms.
- Implementing robust and resilient AI models and algorithms that can detect, prevent, or mitigate malicious attacks, such as using data validation, filtering, and sanitization, applying adversarial training, differential privacy, or federated learning, designing interpretable and explainable AI, and incorporating security metrics and indicators.
- Establishing a clear and transparent governance and accountability structure for AI systems, such as defining roles and responsibilities, setting policies and procedures, ensuring compliance with ethical and legal requirements, and creating mechanisms for oversight, monitoring, and reporting.
- Raising awareness and building capacity among the users and operators of AI systems, such as providing training and education, promoting a security culture, fostering collaboration and information sharing, and engaging with relevant communities and stakeholders.

Performance Metrics, Validation, and Continuous Improvement

A central methodological pillar is the rigorous and ongoing evaluation of model performance and system reliability. Standard metrics, including in particular accuracy, sensitivity (recall), precision, and false positive rate are tracked both continuously and at defined intervals to provide a comprehensive and up-to-date view of each algorithm's effectiveness in detecting unsafe events. The evaluation framework systematically incorporates the use of the confusion matrix, a foundational tool in machine learning that quantifies true positives, false positives, true negatives, and false negatives for every use case. This approach enables a detailed analysis of model behavior, revealing not only overall performance but also highlighting specific areas where improvements may be required or where biases could emerge.

To ensure that models remain fit for purpose in dynamic operational contexts, periodic checks and benchmarking exercises are conducted for each model and use case. These scheduled reviews compare current detection performance against historical baselines, making it possible to identify both incremental improvements and any signs of performance degradation that might result from evolving site conditions, changes in operational patterns, or the phenomenon of domain shift. When such shifts are detected, targeted retraining and recalibration activities are triggered, incorporating recent data and supervisor feedback to restore or enhance model reliability.

This continuous loop of monitoring, evaluation, and improvement not only preserves the validity of event detection but also strengthens confidence among users and stakeholders. By maintaining transparent records of model performance, including confusion matrix snapshots and trend analyses over time, the system ensures that safety analytics are not only robust in theory, but demonstrably effective in the field. The overall accuracy of the models for PPE compliance, Exclusion Zones and Red Zone Monitoring is 85 %. For the following use cases, the achieved AI performance metrics are presented, calculated based on the confusion matrix generated during performance testing:

- Exclusion Zone: % Precision = 95%, Sensitivity = 65%, Accuracy = 78%, and False Positive Rate= 5%
- Red Zone Monitoring: % Precision = 100%, Sensitivity = 52%, Accuracy = 87%, and False Positive Rate = 0%
- PPE non-compliance (helmet): % Precision =100%, Sensitivity 89%, Accuracy = 91%, and False Positive Rate = 0%
- PPPE non-compliance (coverall): % Precision = 100%, Sensitivity = 20%, Accuracy = 79%, and False Positive Rate = 0%

These indicators provide a clear, objective basis for assessing the quality of event detection and the impact of interventions. Trend analysis enables early detection of performance drift, while comparison across sites and event types support targeted retraining and resource allocation. The validation process is not a static checkpoint, but an ongoing dialogue between digital tools and operational reality. Supervisors play a pivotal role in shaping the evolution of the system, with their feedback serving as both a corrective and a creative force. This approach ensures that the platform remains tightly aligned with the changing realities of the business, capable of adapting to unsafe acts and unsafe conditions, changing regulations, and evolving cultural expectations.

The modular architecture of both platforms, local and global supports this adaptability, allowing for the integration of new use cases, analytical features, and reporting capabilities as needs arise. The central tenet remains the same: technology should empower people, inform decision-making, and build organizational resilience, never distract from the fundamental requirement of human vigilance and responsibility.

Conclusion

Throughout its design, deployment, and ongoing operation, the system has been guided by realism, responsibility, and close alignment with field experience. It is not positioned as a turnkey solution capable of eliminating workplace incidents, but rather as a robust enabler, supporting more effective supervision, fostering continuous learning, and enhancing organizational agility.

Managing domain shifts and operational complexity requires transparent communication, disciplined governance, and a strong commitment to adaptability. The system addresses challenges posed by visual ambiguities, environmental variability, and evolving worksite conditions through an iterative cycle of feedback, collaborative review, and agile improvement. This dynamic process ensures the platform remains responsive to real-world conditions, supporting early identification of unsafe acts and unsafe conditions and rapid refinement of detection models.

The integration of AI-powered video analytics and the global dashboard represents a significant advancement in the digital transformation of HSE management. By bridging the gap between granular field observation and strategic insight, this layered architecture delivers actionable intelligence that is both validated and contextually meaningful. The process of standardized event detection and validation enhances data quality and consistency, empowering evidence-informed decision-making at every organizational level.

Through the combination of advanced technology, rigorous operational validation, and strong user engagement, the organization is positioned at the forefront of HSE innovation, ready to address current risks and adapt to future challenges with confidence and integrity.

Acknowledgments

The authors wish to express their sincere gratitude to all field operations, construction, and drilling teams at Saipem, whose dedication to safety and spirit of collaboration have been essential to the success of this initiative. Special thanks are extended to the HSE and subject matter experts for their technical insight and critical contributions throughout the development and validation phases.

Terms & Abbreviations

Term / Abbreviation	Definition
Accuracy	Overall percentage of correct predictions made by the model (true positives and true negatives). Formula: TP+TN/ (Test Set)
Computer Vision	Field of artificial intelligence (AI) focuses on interpreting and extracting information from visual data (images, videos).
Confusion Matrix	Matrix used to evaluate model performance by comparing predicted and actual outcomes (TP, FP, TN, FN).
Dashboard	Digital interface for aggregating, visualizing, and analyzing operational or safety data at site or organizational level.
Event Validation	Process in which detected events are reviewed and confirmed by authorized personnel prior to official reporting or analytics.
False Negative (FN)	Failure of the algorithm to detect an object or event that is actually present.
False Positive (FP)	Incorrect identification by the algorithm of an object or event that is not present.
False Positive Rate	Percentage of negative cases incorrectly flagged as positive by the model. Formula: FP/ (FP+TN)
GDPR	General Data Protection Regulation
HSE	Health, Safety & Environment
HSE Event	Notification or alert of a potential HSE violation generated by the video analytics system (VAWS).
PPE	Personal Protective Equipment;
Precision	Percentage of predicted violations/events that are correct (reliability of positive predictions). Formula: TP/(TP+FP)
Sensitivity (Recall)	Percentage of actual violations/events correctly identified by the model (true positives rate). Formula: TP/ (TP+FN)
True Negative (TN)	Correct identification by the algorithm that no object or event is present when it is absent.
True Positive (TP)	Correct detection by the algorithm of an object or event that is actually present.
Use Case	A defined scenario describing the application of software or technology to achieve a specific goal.
VAWS	Video Analytics for Workplace Safety